22/04/25

IPE (final)

Scheduling

FCFS — first come, first served

SPT — Shortest processing time

EDD — Earliest due date

LPT — Longest processing time

{job/order/work}

Job	Job work time	Job due date
A	6	8
B	2	6
C	8	18
D	3	15
E	9	23

total 28

order same থাকবে।

যা sequence দেয়া থাকবে

FCFS

যেদিন শুরু করা হবে work করে যতদিন লাগবে তত তম দিন

job lateness or tardiness → due date এর পর যতদিন late হবে।

Flow time	Job lateness
6	0
8	2
16	0
19	4
28	5
77	11

B job শুরু হয় 6th day তে work. processing 2days so flow time 8

6th day তে delivery date but flow time 8 so lateness 2

$$\text{Average completion time} = \frac{\text{Sum of total flow time}}{\text{number}}$$

$$\text{Utilization} = \frac{\text{Total Job work time}}{\text{Sum of total flow time}}$$

$$\text{Average number of jobs in the system} = \frac{\text{Sum of total flow time}}{\text{total job work time}}$$

$$\text{Average job lateness} = \frac{\text{Total late days}}{\text{total num of jobs}}$$

example

SPT : Sequence — B-D-A-C-E

small to high processing time

Job	Processing time	Flowtime	due date	lateness
B	2	2	6	0
D	3	5	15	0
A	6	11	8	3
C	8	19	18	1
E	9	28	23	5

delivery date

due date কম হলে sequence এ first এ বসাবে।

processing work time largest সেটা sequence এ first

LPT

delivery date 6, 2 দিন বাড়তি তাই lateness 0

FCFS — customer satisfaction ↑ factory loss ↓,

SPT — comparatively better — company profit ↑ utilization ↑

Critical Ratio (CR)

is a index number found by dividing the time remaining until the due date

$$CR = \frac{\text{time remaining}}{\text{processing time}} \rightarrow (\text{due date} - \text{today's date})$$

CR কম হলে sequencely first এ :

todays date = 0

Job	work time	duedate	CR
A	6	8	$(8-0) \div 6 = 1.33$
B	2	6	$(6-0)/2 = 3$
C	8	18	$(18-8)/18 = 2.25$
D	3	15	$(15-0)/3 =$
E	9	23	$(23-0)/9$

for B, C, D, E work todays date = 6

B $(6-6)/2 = 0$

C

D

E

today date change হতে থাকবে

A	C	D	E	B

Johnson's rule (for two or multiple jobs)

→ to minimize total production time and idle time

Job	Work time for one mach.	Work time for another machine
A	~~5~~	~~2~~ ✓ lowest (right)
B	~~3 second low (left)~~	~~6~~
C	8	9 right
D	10	7
E	7	12

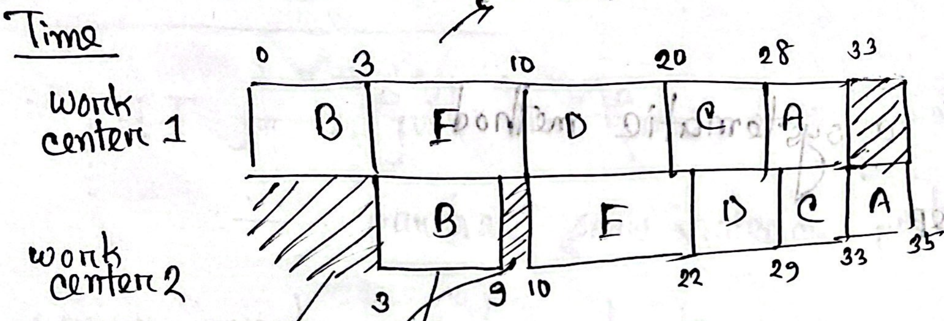
same but min value matter

for sequencing :

- lowest value work center 2 তে থাকলে right এ বসবে এবং ঐ row টা কেটে দিতে হবে।
- then second lowest value center 1 এ থাকলে left এ বসবে

B	E	D	C	A

value same হলে job sequence priority pabe work processing day 7 so (3+7) = 10



• job first এ work center 1 এ যাবার then center 2 তে যাবে।

idle time

WC-2 তে B এর work time 6. so (3+6) = 9

work center-1 এ প্রোসেস করে work center-2 তে যাবে তাই after 03 days start হয়েছে।

Fig : Gant Chart